

Neuroprotective effects of citrus flavonoids.

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Recent attention has been given to the influence of dietary factors on health and mental well-being. Oxidative stress is associated with many diseases including neurodegenerative disorders. Dietary flavonoids exert cardioprotective, chemopreventive, and neuroprotective effects. The biological activities of flavonoids have been attributed to their antioxidant, anti-inflammatory, and signaling properties. A clear understanding of the mechanisms of action, as either antioxidants or signaling molecules, is crucial for the application of flavonoids as interventions in neurodegeneration and as brain foods. Citrus flavonoids exert little adverse effect and have low or no cytotoxicity to healthy, normal cells. The main citrus flavonoids can also traverse the blood-brain barrier; hence, they are promising candidates for intervention in neurodegeneration and as constituents in brain foods. In this review, we discuss the bioactivity, multiple neuroprotection mechanisms, and antioxidant and signaling properties of citrus flavonoids. Receptor-mediated neuroprotective actions and parallel signaling pathways are also explored. Finally, the induction of cellular defense proteins against oxidative stress and neurotoxicity by hesperetin, a main and widespread citrus flavonoid, are also discussed. It is suggested that citrus fruits, which are rich in abundant sources of hesperetin and other flavonoids, are promising for the development of general food-based neuroprotection and brain foods.